

Cashing in On Culture: Local Employment Effects from Art and Cultural District Designation*

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Abstract

U.S. State-designated cultural districts are modern place-based policies that leverage local arts and culture to foster economic growth. While their use is increasing, with more than 375 extant in eighteen states, large-scale evaluations of their efficacy are scarce or non-existent. This study, which relies on geocoded establishment data, finds a modest but persistent increase in employment for some in-district establishments relative to same-industry establishments located beyond district boundaries. The most effective district-implemented incentives for expanding employment appear to be direct grants to establishments, while sales tax credits also show potential benefits.

Keywords: cultural districts, arts, economic development, place-based policies

JEL Classification Numbers: R3, R58

*We thank officials from the Iowa Department of Cultural Affairs, Louisiana Office of Cultural Development, Maryland State Arts Council, New Mexico Economic Development Department, and Texas Commission on the Arts for providing us with cultural district boundaries and for answering numerous questions. We are grateful to Gary Ferrier, Russell S. Sobel, Peter T. Calcagno, Christopher Mothorpe, Norman Maynard, and seminar participants from the College of Charleston, North Dakota State University, Public Choice Society, International Atlantic Economic Society, and Association for Private Enterprise Education for providing many valuable suggestions. We also thank two anonymous referees for several suggestions that have strengthened the paper. Finally, we thank Clara Mantuax for research assistance in digitizing cultural district boundaries. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Any errors are our own.

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1 Introduction

United States (U.S.) state and local governments direct upwards of \$100 billion annually to place-based policies aimed at promoting economic development in targeted areas (Liu and Parilla 2018). Although variation exists across programs, the broad objective is to encourage revitalization and development in geographic areas that are economically disadvantaged and/or historically excluded from public investments. The use of place-based policies is expanding rapidly, having tripled in value since 1990 (Bartik 2017).

Federal empowerment zones (Ham et al. 2011; Busso et al. 2013; Hanson 2009) and state enterprise zones (Neumark and Kolko 2010a; Kolko and Neumark 2010; Billings 2009; Freedman 2013) are two of the better-known place-based policies in the U.S. Like other place-based initiatives, these programs typically target areas characterized by high poverty and unemployment and utilize various benefits, including employment tax credits, investment subsidies, and grants, to encourage economic development and reduce unemployment. However, evidence of their efficacy is mixed (Hanson 2021; Neumark and Simpson 2015).

The creative and cultural economies have gained prominence as viable avenues for promoting and supporting local and regional development. According to a recent report by the United Nations, the cultural and creative sectors account for 3.1% of global GDP (UNESCO 2022). In the U.S., estimates from the Arts and Cultural Production Satellite Account (ACPSA), produced in partnership by the National Endowment for the Arts (NEA) and Bureau of Economic Analysis (BEA), show that art and cultural activities directly account for 4.4% of GDP (\$1,016 billion) and nearly 4.9 million jobs.¹

U.S. state-designated cultural districts are a modern variant of place-based policies that aim to leverage the creative and cultural sectors. These districts provide a highly adaptable policy instrument for local authorities to leverage an area’s artistic and cultural amenities to foster economic growth and development. As evidence of their growing appeal, 18 U.S. states have designated more than 375 districts, with most designations occurring in the past decade.

While most studies focus on a single cultural district or urban area, large-scale evaluations of their efficacy are scarce. This study leverages the National Establishment Time-Series (NETS)

¹These figures are from the 2021 ACPSA estimates. https://nasaa-arts.org/nasaa_research/creative-economy-state-profiles/ (accessed: 3/15/2023).

database, a rich national source of geocoded microdata of U.S. establishments, to investigate the effect of state-designated cultural districts on local labor markets. Using district boundary maps that determine benefit eligibility for financial incentives and a difference-in-differences approach that adjusts for staggered treatment timing, we compare outcomes for eligible establishments inside cultural districts to same-industry establishments just outside district borders.

Our empirical sample includes 188 cultural districts across five U.S. states that offer tax credits or direct grants to individual establishments based on district designation – Iowa, Louisiana, Maryland, New Mexico, and Texas. Districts in these five states account for more than half of all state-designated districts nationwide. We capitalize on industry detail in the NETS database to study the effects of district designation on overall (district) employment, employment in arts and cultural establishments, employment in eating and drinking establishments, retail establishments, lodging establishments, and the formation of new business establishments.

Because cultural district boundaries tend to be delineated around existing cultural assets in areas with underperforming economic indicators, we utilize multiple quasi-experimental designs to mitigate concerns from negative selection. Specifically, we use both difference-in-differences and geographic regression discontinuity strategies that allow for varying degrees of potential spillover effects to establishments located outside district boundaries.

We find evidence of a modest but persistent increase in establishment employment in the arts and cultural sector. Specifically, the average establishment in the arts and cultural sector, classified following Arts and Cultural Production Satellite Account guidelines, experiences between a 2 and 4% increase in employment relative to cultural industry establishments located outside district boundaries. We find no evidence that district designation stimulates new business formation or spills over boundaries, implying the *existing* arts and cultural establishments capture the full program benefits. In terms of specific incentives, direct grants to establishments, and, potentially, sales tax credits for purchases of unique pieces of art, are the most effective at expanding employment.

2 Background

2.1 Evidence on the Efficacy of Place-Based Policies

A large body of research is devoted to understanding place-based policies and their role in shaping economic outcomes. Recently, for example, the U.S. Tax Cuts and Jobs Act of 2017 created

opportunity zones, a place-based policy that allows investors to defer capital gains for long-term investments in distressed areas (e.g., see Chen et al. 2022; Frank et al. 2022; Alm et al. 2021; Freedman et al. 2021). Many U.S.-based studies have focused on empowerment zones and state enterprise zones.² Both programs aim to leverage a variety of benefits (e.g., tax credits, investment subsidies, grants, etc.) to encourage establishments to operate, expand, and hire individuals within the zone’s boundaries, thereby encouraging economic growth and revitalization. While these programs have similarities, empowerment zones are part of a federal program and have greater uniformity in scale, scope, and benefits across states. Meanwhile, because enterprise zones are state-administered programs, there is generally more variation in terms of the mix of benefits, resources, and implementation (Busso and Kline 2008). The evidence on the efficacy of these programs is mixed at best.

For example, Neumark and Kolko (2010a) use the National Establishment Time-Series (NETS) data to study how California’s enterprise program affected employment and the number of establishments. They find little evidence of an effect on either outcome.³ In a related study, Kolko and Neumark (2010) find evidence of heterogeneous effects from California’s enterprise program depending on implementation and administration. Specifically, they find that zones with greater marketing and outreach efforts (as they relate to survey responses by zone administrators) are more likely to show evidence of positive employment effects.⁴

Billings (2009) utilizes a matched business establishment sample to explore whether Colorado’s enterprise zones influence local economic activity. His results suggest that establishment location decisions are seemingly unaffected by the enterprise zone’s incentives. However, he does find evidence of a positive employment effect in some specifications. This is consistent with Freedman (2013), who finds that enterprise zones increased employment in Texas after taking into consideration the

²Examples of research exploring the effect of other place-based initiatives on various economic outcomes include Kitchens and Wallace (2022), Freedman (2012), Freedman et al. (2023), Harger and Ross (2016), Been et al. (2016), Meltzer (2012), Theodos et al. (2022), Lu et al. (2019), Franco and Macdonald (2018), Fernandez and Martin (2020) and Koster and Rouwendal (2017).

³Some of their results suggest that the number of establishments fell. Neumark and Kolko (2010a) argue that a fall in the number of establishments without any discernible effect on employment implies that establishments are becoming larger, which may suggest that enterprise zone benefits are more attractive to larger firms but not to smaller ones (perhaps due to the administrative burden).

⁴They do caution, “however, that we are measuring zone managers’ own perceptions of local marketing and outreach efforts. It is possible that managers view themselves as more active in marketing and outreach after observing positive employment effects in their zones, in which case the self-assessment of marketing activity could be the result of employment growth rather than a contributor” (Kolko and Neumark (2010), pg. 34).

fact that the zones are assigned to census block groups meeting a particular poverty threshold.

Work by Ham et al. (2011) provides evidence of positive employment and wage effects stemming from federal empowerment zones, enterprise community programs, and state enterprise zones programs.⁵ Busso et al. (2013) also find positive employment effects from the federal empowerment program, but Hanson (2009) finds no evidence of a change in employment or poverty after instrumenting for empowerment zone designation.

2.2 State-Designated Arts & Cultural Districts

The term cultural district can be broad and refer to formal and informal (or organic) efforts to recognize a geographic area as a “hub” of cultural activity, given the concentration of arts or cultural amenities in that area (Redaelli 2019; Frost-Kumpf 1998; Galligan 2008; Gallagher 2021; Markusen and Gadwa 2010a; Chapain and Sagot-Duvauroux 2020). In this paper, we limit our analysis to *state*-designated cultural districts, which formalize the state’s role in the designation or certification process.^{6,7} (Program nomenclature also varies across states, so we use the term “cultural district” as the inclusive descriptor for state-designated art, entertainment, and cultural districts.)

While state-designated cultural district programs tend to be smaller in scale and scope than other place-based policy programs like the ones discussed in the preceding section, their use has grown rapidly since the early 2000s. These districts are one example of “creative placemaking” (Markusen and Gadwa 2010b; Markusen 2014) initiatives that aim to use local arts and culture to promote economic development.⁸ In particular, state-designated cultural districts promote these

⁵Neumark and Young (2019) show that many of the positive effects estimated by Ham et al. (2011) are no longer significant after considering pre-intervention differences.

⁶Ferreira Neto (2021) investigates the determinants that influence the diffusion of cultural district programs in the USA. His findings suggest that government competition and imitation help explain the adoption (or formalization) of cultural district laws.

⁷Other states, like Maine, Missouri, Ohio, Virginia, and Wisconsin, empower municipalities to create local districts, but lack statewide recognition or roles for state agencies in the process (Ferreira Neto 2021; National Assembly of State Arts Agencies 2020). While our focus is within the U.S. context, similar recognitions or designations have been used, for instance, in Europe (Chapain and Sagot-Duvauroux 2020). Le Blanc (2010) and Mizzau and Montanari (2008), for example, examine the effects of cultural districts in different regions of Italy.

⁸According to the National Endowment for the Arts, “Creative placemaking integrates arts, culture, and design activities into efforts that strengthen communities. Creative placemaking requires partnership across sectors, deeply engages the community, involves artists, designers and culture bearers, and helps to advance local economic, physical, and/or social change, ultimately laying the groundwork for systems change. This definition is intentionally open and broad because creative placemaking draws on all artistic disciplines, and can be deployed as a strategy to address a wide range of community issues or challenges from public health to safety, economic development to housing” (<https://www.arts.gov/impact/creative-placemaking>; accessed: 7/15/2022). Thus, creative placemaking serves as a term intended to reflect how place-based arts and cultural policies can foster local economic development.

areas by leveraging the arts and cultural institutions that characterize those locations to attain various goals including: attracting artists and cultural enterprises to the area, encouraging business and job development, establishing the area as a tourist destination, promoting the preservation and reuse of historic buildings, enhancing property values, and fostering local cultural development (National Assembly of State Arts Agencies 2020).

The end goals for cultural district programs can be quite similar, however, their implementation can vary across programs since these are independently established across states. Typically, the designation process is initiated by a local authority that submits an application to have a (contiguous) zone or area recognized as a hub of cultural activity (Noonan 2013; Frost-Kumpf 1998; Redaelli 2019). A description of the key cultural anchors and resources characterizing the area, which could include a high concentration of cultural/artistic enterprises, facilities, and/or individuals, as well as other related activities like events and festivals, is an important justifying component of the application. A description of the anticipated benefits, goals, and promotion and development strategies are generally necessary as well. The pertinent state authority then reviews and evaluates the application to decide whether the district designation will be granted.

If the designation is granted, programs are similar in that artists and establishments within district boundaries become eligible for various benefits, including access to technical expertise and financial assistance in the form of grants or tax credits. The available resources for state designation and operation of the district, as well as the composition of benefits available for eligible establishments, are likely state-dependent given the local fiscal and political environment.⁹

States with formal programs for designated cultural districts are shown in Table 1. Except for Kentucky and Oklahoma, the programs were created through legislation. The programs in Kentucky and Oklahoma are administered by the state arts agency, which creates guidelines, adjudicates applications, designates districts, and monitors operations. They are therefore operationally equivalent to any other state program authorized by the legislature.

[Table 1 here]

The first state to establish a formal recognition program was Rhode Island in 1998. Table

⁹An income tax credit, for instance, would be ineffective in a state like Texas which has no income tax. Other states may have no incentives (or a cultural district program) in place due to the fiscal considerations (National Assembly of State Arts Agencies 2020).

1 reports the number of cultural districts designated in each state and their program’s inception (through April 2022), with Louisiana having the most districts (118). Of the 18 (state-designated) cultural district programs, only six offer direct financial assistance to individual establishments; Iowa, Louisiana, Maryland, New Mexico, and Rhode Island offer tax credits, while Texas offers grants. States that do not offer financial incentives directly to establishments provide other incentives such as (i) formal recognition of the area as a cultural hub, (ii) technical assistance such as marketing and advertising, meetings, and networking, or (iii) district-level grants to support general operations.

In terms of raw counts, Louisiana has certified more districts than other states, possibly for several reasons. First, south Louisiana is the historical home of the Cajuns and Creoles, whose culture is very distinct in the United States. The majority of Louisiana’s certified districts are located in the southern part of the state. Additionally, Louisiana was one of the early adopters of cultural districts, enacting authorizing legislation in 2007. This has given local governments more time to observe how districts operate in practice and to propose new districts. Ironically, on a per year basis, Texas certified more districts (51) than Louisiana in its first seven years after the authorizing legislation was approved.

Interest in the role culture, cultural amenities, and cultural districts play in economic development is not new (e.g., Boualam (2014), Bereitschaft (2014), Markusen and Gadwa (2010a), Lazzeretti (2003), Frost-Kumpf (1998), Santagata (2006), Santagata (2011), Schuetz (2014), Currid and Williams (2010), Gutierrez-Posada et al. (2022), Tao et al. (2019) and Florida and Mellander (2009)). Schuetz (2014), for instance, finds that art galleries in New York locate in high-amenity neighborhoods but do little to stimulate redevelopment. Similarly, recent work on film incentives, such as Button (2019), Bradbury (2020), and Button (2021), finds scant evidence that these programs expand employment or business formation.

Quantitative investigations of the efficacy of U.S. state-designated cultural districts are relatively scarce (National Assembly of State Arts Agencies 2017; Gallagher 2021). In an early study, Brooks and Kushner (2001) use regression analysis on a sample of six cities over 17 years. They use a dummy variable to identify when a cultural district is implemented and find a negative association between cultural districts and economic growth. More recently, Noonan (2013) find that cultural districts have a positive effect on median property values, employment, and income using fixed-

effects regressions on neighborhood-level census data. Like this paper, Portillo and Wagner (2021) uses a difference-in-differences approach that adjusts for staggered treatment timing to estimate the effect that district designation has on residential real estate prices. Using individual property sales in Louisiana, they find that residential properties inside cultural district boundaries sell for 7% more post-designation relative to residential properties outside the districts. Finally, Breznitz and Noonan (2018) explore how cultural districts and art education programs affect arts, culture, and media-related jobs using zip-code level data. Their regression results show some evidence that cultural districts can exert a positive effect on employment, however, they conclude that proximity to arts and digital media education programs may be more important for expanding employment than proximity to cultural districts.

Building on this work, we seek to provide a more comprehensive evaluation of the role state-designated cultural districts play in shaping local economic labor market outcomes, a predominant objective in the United States.¹⁰ Because programs that provide direct incentives to establishments may be more likely to affect local outcomes, we choose to focus on them.

Table 1 provides summary information on each program’s financial incentives. Maryland’s program specifically exempts “the admissions and amusement tax gross receipts from any admissions or amusement charge imposed by an ‘arts and entertainment enterprise’ or a ‘qualifying residing artist’ in an A&E District.”¹¹ Maryland also offers local property tax incentives, whereby “The owner of a manufacturing, commercial, or industrial property is eligible for a local property tax credit based on the difference between the pre- and post-renovation assessments of the property when it has been wholly or partially renovated for use by ‘qualified residing artists’ or ‘arts and entertainment enterprises.’”

Programs in Iowa, Louisiana, and New Mexico utilize preservation tax credits, designed to incentivize the rehabilitation of historic residential and commercial buildings. Although details and eligibility vary, preservation credits are generally limited to a percentage of qualified rehabilitation

¹⁰State-designated cultural districts may also generate benefits beyond economic ones. In particular, by formally recognizing or certifying an area as a cultural district, local authorities pay homage to what makes that area unique and different (Gallagher 2021; Galligan 2008; Frost-Kumpf 1998). Furthermore, they utilize those local anchors as the catalyst for revitalization, thereby promoting and enhancing the intrinsic values of the local arts and culture. Even though these programs are independently established across states, a common thread across them is the expectation that state designation will enhance a sense of place and community, as well as promote public-private partnerships for further development.

¹¹<https://www.msac.org/programs/arts-entertainment-districts> (accessed: 07/19/22).

expenditures (QREs).¹² These credits can generally be combined (or “stacked”) with the federal historic tax credit to provide an even larger total incentive. Unlike Maryland’s local property tax credit, which is limited to specific occupations or organizations, all business organizations or individuals with property located inside cultural districts are eligible for preservation credits.

Several programs also offer sales or income tax credits. For instance, Maryland and Rhode Island offer income tax credits that enable artists to lower their tax liability on income derived from arts and cultural activities, such as sales and performances. Similarly, Louisiana and Rhode Island exempt the sale of original art within district boundaries from state and local sales taxes.¹³ Rhode Island passed legislation in 2013 that extended the sales tax exemption for works of art “to all artists and galleries working in Rhode Island, *effectively making the entire state a cultural district*” (emphasis added) (National Assembly of State Arts Agencies 2020). Since we cannot observe counterfactual outcomes for ineligible Rhode Island businesses after 2013, we omit the state from our analysis.

The final state in our sample, Texas, incentivizes businesses via direct grants. That state’s program awards grants ranging from \$3,000 to \$250,000 directly to arts organizations located within district boundaries. Eligibility is restricted to arts organizations, arts organizations of color, local government arts agencies, rural arts providers, and college arts institutions. Recipients must also generate at least \$50,000 in revenue and agree to provide matching expenditures.¹⁴

Other state programs award grants to the local authority overseeing district operations rather than individual establishments. While organizations or artists may benefit from these grants, any effect is indirect. As one example, Massachusetts offers a Cultural District Investment Grant where managing entities may receive grants of \$7,500 (with no matching requirement) to support district operations.¹⁵

¹²Preservation credits in Iowa and Louisiana are limited to 25% and 20% of QREs, respectively. New Mexico’s program has no minimum QRE and a maximum benefit of \$50,000. Iowa’s and Louisiana’s preservation credits also have minimum expenditure requirements for eligibility. Iowa and Louisiana cap their annual aggregate preservation tax credits at \$45 million and \$125 million, respectively. New Mexico’s preservation credit has no aggregate cap.

¹³Louisiana eliminated its state sales tax exemption in July 2018. However, a local tax exemption remains in place.

¹⁴Information on Texas’s grant program, including past recipients, can be found here: https://www.arts.texas.gov/wp-content/uploads/2018/01/Cultural_District_Funding_TCT.2017.pdf. Program guidelines are available here: <https://www.arts.texas.gov/ow/tcagrants/TXArtsPlan/CDP.htm> (accessed 7/21/22).

¹⁵<https://massculturalcouncil.org/communities/cultural-districts/cultural-district-investment-grants/application-process/> (accessed: 7/27/22). District-level grants are also a feature of other programs, such as those in California, Colorado, Maryland, and Oklahoma (National Assembly of State Arts Agencies 2017).

In our empirical analysis, however, we limit the focus to state-designated programs that provide direct financial incentives to individual establishments. Figure 1 highlights the five state programs in our sample and shows the approximate location of individual cultural districts within each state.

[Figure 1 here]

Specific goals of the programs in our sample are reported in Table 2. Although cultural district objectives may vary from place to place or country to country, economic revitalization has been a primary objective of districts located in the U.S. For example, Iowa’s Cultural and Entertainment Districts (CEDs) have goals to “attract residents and tourists who support adjacent businesses such as restaurants, lodging facilities, retail stores and entertainment venues, enhancing property values, the profitability of surrounding businesses and the tax base of the area.”¹⁶ The program objectives serve as a guide for the empirical analysis outlined in the following sections.

[Table 2 here]

3 Data

Since cultural districts generally enclose areas where art and cultural assets are already located, they tend to be small and not coincide with conventional boundaries where economic measurement occurs, such as census blocks or tracts. Some cultural districts are completely subsumed within a single census block, while others overlap multiple blocks. Consequently, errors would be likely if district-level outcomes were constructed using conventional data sources, such as the American Community Survey, which does not provide the precise locations of business establishments.

Cultural district programs often aim not only to encourage job or business creation, but to foster cultural development, promote artists and artisans, boost real estate values, and increase tourism. While examination of multiple objectives might yield a more complete picture of benefits, we are limited by the rarity of data. Published reports on cultural districts often report statistics like the number of events, estimates of attendance and spending, or artist sales *after* certification. Unfortunately, these data were rarely (if ever) tracked within relevant boundaries *prior* to certification.

¹⁶<https://culture.iowaeda.com/iowa-cultural-entertainment-districts/> (accessed: 1/20/2024).

Therefore, we test the effect of cultural district certification on observable outcomes of interest – establishment counts and employment levels. Improving local economic conditions, which includes job or business creation, is an important aspect of program goals (see Table 2).

We form cultural district-level economic outcomes using the National Establishment Time-Series (NETS) database. This database was developed by Walls & Associates, LLC, and is a private-sector alternative to the Census Bureau’s restricted-use Longitudinal Business Database. The NETS data are derived from Dun and Bradstreet’s (D&B) collection of Duns Marketing Information files that track more than 71 million establishments annually (in January) (Walls & Associates 2019). D&B tracks individual establishments to create, and ultimately sell, marketing and credit scores. Information is collected from public sources such as court records and secretaries of state, as well as direct contacts by telephone (Barnatchez et al. 2017). Every establishment is assigned a unique identifier – a DUNS number – so it can be tracked over time. D&B has a strong reputation for accuracy. For example, to validate legitimate businesses, every business establishment that applies for federal government contracts or financial assistance is required to include their (unique) DUNS identifier on their U.S. General Services Administration application.¹⁷

In addition to providing the latitude and longitude of establishments, the NETS database includes counts of employees, whether the establishment has relocated, the first and last year of active operation, D&B’s proprietary PAYDEX credit score, and the establishment’s primary 6-digit North American Industry Classification System (NAICS). The detailed NAICS codes are particularly valuable in our setting, as they enable us to explore how district certification affects different industries.

Because the spatial distribution and concentration of the cultural value chain is interconnected across sectors (Dellisanti 2023; Chapain and Sagot-Duvauroux 2020), we examine how district certification affects total district employment. If certification enhances economic development and stimulates creativity and spillovers, any effect may be visible by examining aggregate employment and establishment counts.

Next, districts are mixed-use areas with high concentrations of arts and cultural amenities, which serve as the anchors of attraction (Frost-Kumpf 1998). Therefore, many of the arts and cultural

¹⁷In 1998, the federal government codified the use of the DUNS identifier in the Federal Acquisition Regulations. The provision to include a DUNS number was changed in April 2022 because the U.S. now requires establishments to obtain a Unique Entity Identifier (UEI) created by SAM.gov.

establishments within district boundaries will be eligible for the program’s financial incentives. Using each establishment’s NAICS code, we explore how certification affects employment in the “core arts cultural sector,” as defined by the Arts and Cultural Production Satellite Accounts (ACPSA). The included sectors, shown in Appendix Table A1, range from artists and traditional performing arts and museums to landscape, graphic, and computer systems design.

Defining and measuring cultural activity, cultural industries, or creative placemaking, often depends on context and a project’s larger scope or objectives. We opted to rely on the ACPSA definition for several reasons. One, as noted by the Bureau of Economic Analysis, arts and cultural production in the ACPSA is defined “to be largely consistent with definitions used by the United Nations and the European Union” (Kern et al. 2014). A key goal was to use input-output relationships in the broader economy and capture the “creative chain” of economic activities associated with art and cultural production (Kern et al. 2014). Most importantly, the ACPSA maps the creative chain to the North American Industrial Classification System, precisely matching how establishments are categorized in the NETS database. This enables us to identify cultural establishments using a rigorous structured system rather than an ad hoc process.

Since districts also aim to enhance a sense of place and attract more visitors, we separately investigate how certification affects employment and establishments in sectors that may offer complementary activities to district residents and visitors. These include: (i) eating and drinking establishments (NAICS 722), (ii) retail establishments (NAICS 45) that sell merchandise to the public for personal or household consumption, and (iii) hotel/motel establishments where visitors could stay (NAICS 722110). The retail trade sector (NAICS 45) is broad, encompassing health and personal care stores, food and beverage stores, and clothing and general merchandise stores, to name a few. If certification increases the number of residents and/or visitors to the district, these sectors providing complementary services may be the most likely to experience economic benefits.

NETS data have been used in numerous studies including, but not limited to, business relocations (Neumark et al. 2005), establishment-level job growth (Neumark et al. 2011; Choi et al. 2013), enterprise zones (Neumark and Kolko 2010b), local entrepreneurship (Clayton et al. 2019), business survival (Wagner and Lee 2024), national monument designation (Walls et al. 2020), and trends in business concentration (Rossi-Hansberg et al. 2020).

As Neumark et al. (2011) document using California data, NETS includes more establishments

(and higher employment counts) than conventional government data sources because it includes nonemployer establishments, in other words, sole proprietors with no paid employees. This is an important segment in the arts and cultural sector; the National Endowment of the Arts reports that artists and other cultural workers are 3.6 times more likely to be self-employed than workers in other industries (National Endowment of the Arts 2019), and the NETS database allows us to capture them.¹⁸

Individual district boundaries (or shapefiles) were obtained from the Department of Cultural Affairs (Iowa), Office of Cultural Development (Louisiana), State Arts Council (Maryland), Economic Development Department (New Mexico), and Commission on the Arts (Texas). Since our NETS sample ends in 2019, districts certified after 2019 are excluded. The earliest districts in our sample were certified in 2002. Between 2009 and 2019, an average of 15 new districts were designated annually, with 17 being certified in 2019.

[Table 3 here]

Table 3 presents descriptive information by state. Although the typical cultural district covers an area of less than 1 square kilometer in every state except Louisiana, there is widespread variation. The smallest district among our states, measuring only 0.02 square kilometers, is located in Denton, TX, a city in the Dallas-Forth Worth metropolitan statistical area. In contrast, the Bayou des Glaises cultural district, located in Avoyelles Parish, Louisiana, envelops more than 180 square kilometers.

Treated establishments are located within a state-designated cultural district boundary, while comparison establishments are outside a district’s boundaries (within 1600 meters) and ineligible for incentives. Nearby establishments may provide a suitable counterfactual for treated establishments, as they likely share unobservables that could affect economic outcomes, such as public transportation access or local zoning ordinances.

The average district has between 152.4 establishments (New Mexico) and 1238.2 establishments (Texas) within its borders, with per establishment employment counts ranging from 6.4 in Louisiana

¹⁸Several studies have validated the NETS data against other official data sources (Zeballos and Marchesi 2022; Rossi-Hansberg et al. 2020; Barnatchez et al. 2017). Barnatchez et al. (2017) finds that while NETS data generally mimic the Quarterly Census of Employment and Wages, there is more imputation among small establishments. See the data appendix of Rossi-Hansberg et al. (2020) for a detailed evaluation of the NETS database.

to 10.6 in Maryland. Amongst comparison establishments, average employment counts vary from 8.3 in New Mexico to 12.7 in Texas. On a per-cultural district basis, roughly 1,250 people work within a typical district in Iowa, 1,300 in New Mexico, 2,500 in Louisiana, 8,200 in Maryland, and 10,900 in Texas. Aggregate district employment ranges from 6,250 in Iowa to more than 510,000 in Texas.

For the typical cultural district, cultural establishments (as defined by the APCSAs) account for 4-6% of all establishments and 4-17% of total employment. Since cultural districts generally encompass small areas, the average number of establishments within one tends to be smaller than the number located within 1600 meters of a boundary.

If cultural districts do spur development, economic activity may spill over district boundaries. We allow for this possibility using several comparison groups based on varying distance thresholds. Because the risk of unobservables increases as the distance between treated and comparison establishments expands, we limit the maximum distance for comparison establishments to 1600 meters (or 1 mile).

4 Empirical Strategy

To estimate the causal effect of cultural district designation on local outcomes, we employ a difference-in-differences identification strategy. In this framework, we compare outcomes for establishments (number of employees, number of establishments) located within cultural districts to those for establishments just outside district boundaries. While state-designated cultural districts tend to target economically distressed areas with a high concentration of cultural amenities, we believe our quasi-experimental approach helps mitigate any concerns from negative selection by comparing establishments with otherwise similar economic conditions before and after certification. To further mitigate this concern we also use a geographic difference-in-discontinuity strategy, discussed in greater detail in Section 5.2.

Our generalized panel form of the difference-in-differences model is given by:

$$Y_{ict} = \alpha + \beta D_{ict} + X_{ct}\theta + RT_t\phi + E_i\nu + \epsilon_{ict} \quad (1)$$

where Y_{ict} is the outcome of interest for establishment i , located in county c , at time t . X_{ct} is a vector of observable, time-varying county characteristics that may be correlated with local

economic outcomes. RT_t denotes the relative time (or time to treatment) fixed effects that will adjust for shocks common to all establishments. Establishment-level fixed effects (E_i) will adjust for time-invariant factors at the establishment level such as the size of a given cultural district.

D_{ict} is the difference-in-differences variable of interest; it equals unity if establishment i is located within the boundaries of a certified cultural district after the district was certified; it equals zero otherwise. β is the difference-in-differences parameter of interest.¹⁹

Individual states enacted programs in different years, and, within each state, districts have been certified at different points in time. Given the limitations of the NETS data, which ends in 2019, individual districts are included if we can observe at least one pre- and one post-certification observation. Naturally, we relax this restriction when we focus on new establishment formation.

We follow the simple approach proposed by Cengiz et al. (2019) to address the fact that districts are certified at different points in time. Specifically, we recenter the data to relative time so that every district is certified at the same point in time (time 0). Relative time period -1 is one year prior to certification, and relative time period +1 is one year after certification. To make this recentering strategy concrete, consider two cultural districts (A and B) certified in 2005 and 2014, respectively. In relative time, district A is treated in period 0 (2005), and we include observations on individual establishments in, or near, district A from -8 (1997) to +8 (2013). For district B, treated in period 0 (2014), we include observations from -8 (2006) to +5 (2019). We are only able to include 5 post-certification periods for district B because of data limitations.²⁰

The Cengiz et al. (2019) recentering strategy ensures that establishments are always compared to “clean comparisons” because it collapses the regression to a single treatment period for all treated establishments (relative time = 0). If there is treatment effect heterogeneity and one *fails* to adjust for the staggered timing of certification, then estimates would be biased because establishments treated “early” in the sample would end up as part of the comparison group for establishments

¹⁹The canonical two-group two-period (2 x 2) difference-in-differences equation includes three indicator variables: (i) a treatment indicator that equals unit for treated units, (ii) a post indicator that equals unity in the post treatment period, and the interaction of (i) and (ii), which captures the average difference between the groups. Since we have repeated observations on establishments for more than two periods, we cannot include the conventional treatment or post indicator variables because they are perfectly colinear with the establishment and relative year fixed effects. Our difference-in-differences variable of interest, D_{ict} , is equivalent to (post * treatment) and can be included because it varies by group and time.

²⁰Because a comparison establishment could be “close” to more than one certified district, we assign establishments to the comparison group for its closest cultural district. This ensures that each comparison establishment appears only once in the sample.

treated “later” in the sample, sometimes called the “forbidden comparison” (Roth et al. 2023). Recentering is a very simple and intuitive way to avoid this concern. Moreover, as Baker et al. (2022) has demonstrated through simulations with heterogenous effects across both groups and times, the Cengiz et al. (2019) approach recovers the same average treatment effects as other estimators, including Callaway and Sant’Anna (2021) and Sun and Abraham (2021).

Relative to the Cengiz et al. (2019) approach, the Callaway and Sant’Anna (2021) estimator does have one advantage in that one can make group-time comparisons for, say, establishments treated in 2010 versus 2014. This would not be very informative in our application because, in any given year, districts are certified across multiple states that utilize different financial incentives. In other words, the cohort of districts certified in 2010 looks very different than the cohort of districts certified in 2012 or 2015, both in terms of location and the types of financial incentives available to cohort members. We do take steps, to the extent possible, to explore the potential for heterogeneous effects in Section 5.3.

To assert a causal claim from a difference-in-differences design, one must be willing to assume that establishment employment in the comparison group after district certification represents a valid counterfactual for how employment would have evolved in establishments located within district borders if the cultural district was never created (i.e. the parallel trends assumption). This identifying assumption is impossible to empirically test because one never observes how employment *would have evolved* in the establishments located within cultural district boundaries if the districts never existed.

Structural differences between treated and comparison establishments are permissible under the parallel trends assumption as long as those differences remain constant during the sample period. Establishment-level fixed effects also play an important role in helping to mitigate confounding local factors that are time-invariant, which could include an establishment’s proximity to other local amenities or disamenities, access to transportation services, or local zoning ordinances.

From a confounding perspective, the greatest threat to identification is time-varying policies that affect a subset of treated or comparison establishments. We include several time-varying county- and state-level covariates to adjust for observable factors that might influence trends in local employment or establishment counts.²¹ The primary variables include: county median household

²¹Sub-county time-varying covariates are not available annually before the 2009 5-year American Community

income, state top marginal income tax rate, county unemployment rate, and the trend growth in personal income in the county over a 10-year period. The 10-year growth rate in personal income will capture longer-term trends in an area’s overall economic performance.²²

Additionally, since the industry composition of establishments located inside and outside of cultural districts is not identical, more labor-intensive establishments could potentially experience a larger employment change from district financial incentives than establishments in more capital-intensive industries. To adjust for this concern, we construct an employment-weighted labor intensity variable to control for differences in factor intensity between treated and comparison group establishments. Using Jorgenson et al. (2019)’s labor intensity estimates, available at the four-digit North American Industry Classification System (NAICS) dimension, group j ’s (treated/comparison) labor intensity measure at time t is the employment-weighted average labor intensity formed from all establishments that constitute each cultural district’s treatment and comparison group.

Next, we construct an establishment-level measure of traveler proximity using airline passenger enplanement data from the U.S. Bureau of Transportation Services. This variable will adjust for the fact that establishments and cultural districts located closer to high-traffic or tourist areas may experience more business, all else equal. Using the latitude and longitude of sample establishments and U.S. airports, we calculate each business establishment’s inverse-distance weighted average passenger enplanements per year to every airport in the United States. We then scale this figure by the county’s population so that establishments with higher traveler proximity scores are located in areas closer to a greater volume of population-adjusted airline traffic.

There is also recent evidence that creative and cultural industry employment is positively associated with the local endowment of creative human capital (Denti 2022). For every establishment in the sample, we construct a measure of the stock of creative human capital using the establishment’s inverse-distance weighted (time-varying) enrollment of liberal arts, music, and theater majors at colleges and universities.²³ All else equal, establishments in closer proximity to more

Survey.

²²County median household income is from the Census Bureau’s Small Area Income and Poverty Estimates program. State marginal income tax rates are from the National Bureau of Economic Research. Trend growth in county personal income is the ten-year growth in nominal personal income (from the Bureau of Economic Analysis), and county unemployment data are from the Bureau of Labor Statistics.

²³Individual college/university data from the National Center for Education Statistics Common Core of Data.

college students enrolled in creative class majors will have higher values of creative human capital.

Finally, establishments may also be eligible for other financial or technical assistance based on their location. We therefore include an indicator variable that equals unity for establishment i at time t if the establishment is located within the boundaries of a federal empowerment zone, federal renewal community, or state or local enterprise zone. These are the three most comprehensive place-based policies designed to boost local economic development.

5 Results

5.1 Employment

We begin by exploring how cultural district certification affects establishment employment. Table 4 presents the difference-in-differences estimates. The dependent variable is the log of the number of employees, and regressions have been estimated with establishment fixed effects, relative time (or time-to-treatment) fixed effects, and a constant term that are not reported. Standard errors, shown in parentheses, are multiway clustered at the cultural district and metropolitan statistical area dimensions.²⁴

As noted earlier, we focus on five establishment groups—all (which captures aggregate employment effects for establishments within cultural districts), cultural (establishments whose operations lay within the arts and cultural sectors), eating and drinking, retail, and hotel/motel establishments (establishments providing complementary services to residents and visitors that may also benefit if more people visit the area by being attracted to the art and cultural amenities highlighted by the district’s certification).

Results for all establishments are reported in columns 1-4, for cultural establishments in columns 5-8, for eating and drinking establishments in columns 9-12, for retail establishments in columns 13-16, and for hotel/motel establishments in columns 17-20. For each regression, we also report the number of distinct treated and comparison establishments.²⁵

Across each group of regressions (All, Cultural, Eating/Drinking, Retail, Hotel/Motel), the

²⁴Since the dependent variable is the log of establishment employment, the average percentage change in establishment-level employment after a cultural district was certified (relative to comparison establishments) is given by $100 * (e^{\beta} - 1)$, where β is the parameter of interest (Giles et al. 2011). For small coefficient estimates, such as those in Table 4, the percentage change is closely approximated with $100 * \beta$.

²⁵The specific NAICS codes defining each sector we explore are noted in Section 3. Appendix Table A1 lists the NAICS codes considered to be “core arts and cultural” by the Bureau of Economic Analysis’s Arts and Cultural Production Satellite Account (ACPSA).

distance threshold and buffer change moving from left to right. Using all establishments to illustrate this point, column (1) uses establishments within 800 meters of a cultural district boundary as the comparison group, while column (2) expands the boundary to 1600 meters. The final two columns use a distance threshold of 1600 meters but exclude establishments “close” to district boundaries to enable exploration for spillover effects (Neumark and Kolko 2010a; Neumark and Simpson 2015; Hanson and Rohlin 2013).

Positive spillovers would be present if district designation increases employment both inside and outside the targeted zone. This would also dampen the true effect of designation. In contrast, if employment relocates from comparison to treated establishments, then this would overstate the effect of designation. For each outcome – All, Cultural, Eating/Drinking, Retail, and Hotel/Motel – the final two regressions omit comparison establishments within 200 and 400 meters of the boundary to enable exploration of these concerns.

We fail to find any discernible employment effect stemming from cultural district designation, relative to the comparison establishments, among all establishments (columns 1-4), eating and drinking establishments (columns 9-12), retail establishments (columns 13-16) or hotel/motel establishments (columns 16-20). The point estimates for all establishments vary between 0.3 and 1.0%, but they are not statistically different from zero.

In contrast, we find evidence that cultural establishments experience an increase in employment between 2.6 and 3.2% relative to their comparison establishments. The estimated coefficients are similar across specifications (i.e., distance and buffer thresholds), suggesting that spillover effects are not a practical concern. Considering that cultural employment ranges between 4 and 17% of district employment, our results suggest that the estimated increase is too small to measurably boost overall district employment.

[Table 4 here]

We also estimate event study formulations to probe the parallel trends assumption holding pre-certification. Figure 2 shows selected plots for all establishments (columns 1 and 4 in Table 4) in the top row, core cultural establishments (columns 5 and 8) in the second row, eating/drinking establishments (columns 9 and 12) in the third row, retail establishments (columns 13 and 16) in

the fourth row, and hotels/motels (columns 16 and 20) in the bottom row. The event study plots that are omitted from Figure 2 do not differ in any meaningful way from the plots we report.

The event study plots reveal no evidence of a statistical difference in employment counts between treated and comparison establishments pre-certification in any sectors, suggesting that comparisons constitute a suitable counterfactual for the treated establishments.

[Figure 2 here]

While the estimates in Table 4 reflect the average post-certification difference across the full sample, the event study plots reveal the dynamic response. One year post-certification, there is a significant difference between treated and comparison cultural establishment employment that persists through the post-treatment sample. There is no evidence the effect is transitory.

5.2 Robustness Check using Geographic Discontinuities

A causal interpretation from the difference-in-differences specifications in Section 5.1 hinges on the validity of the parallel trends assumption. However, since we know the precise location of cultural district boundaries determining tax credit eligibility and the latitude and longitude of treated and comparison establishments, we can explore the robustness of the baseline difference-in-differences estimates using a geographic difference-in-discontinuities approach similar to Grembi et al. (2016) and Koster and Ommeren (2019). An alternative empirical strategy is useful to confirm that the estimated treatment effects we find for cultural establishment employment is not an artifact of a single estimation strategy.

The geographic difference-in-discontinuities is one form of a regression discontinuity design (RDD) that is applied to panel data and uses the distance to an administrative border as the running variable. Because the conventional RDD estimator is designed for cross-sectional data, studies such as Grembi et al. (2016) and Koster and Ommeren (2019) apply the conventional RDD estimator to a first-differenced equation using observations before and after some policy intervention. The resulting estimated treatment effect shares a similarity with difference-in-differences in the sense that one is capturing the difference in the change between treated and comparison groups at the geographic boundary after the policy intervention. If establishments just outside the district boundaries are similar (on average) in many relevant respects to establishments inside the district

that are eligible for financial incentives, then one can assert a causal claim.

For each of the four baseline comparison groups described in Section 5.1, we estimate a first-differenced RDD that uses the establishment’s distance to the nearest cultural district border as the running or forcing variable. We use the same relative time period, -8 to +8, as the baseline difference-in-differences samples, and we include the full set of time-varying covariates in the RDD regressions that appear in the difference-in-differences specifications. Specifically, the difference-in-discontinuity equation can be expressed as:

$$(\bar{Y}_{ic}^{post} - \bar{Y}_{ic}^{pre}) = \beta_0 + \tau D_{ic} + h(W_{ic}) + (X_c^{post} - X_c^{pre})\theta + \varepsilon_{ic} \quad (2)$$

where D_{ic} denotes assignment to treatment, W_{ic} is the running variable, and h is a flexible function in W . W_{ic} is establishment i ’s (located in county c) distance in meters to the nearest cultural district boundary. Treatment occurs at a distance of zero, with negative distances referring to comparison establishments and positive distances referring to treated establishments.

The outcome variable is the *difference* in the mean of establishment i ’s log employment post-certification ($\bar{Y}_{ic}^{post}$) and the mean of establishment i ’s log employment pre-certification ($\bar{Y}_{ic}^{pre}$). Each time-varying county characteristic included in X_c is also transformed into its pre- and post-certification mean difference. Differencing at the establishment level eliminates unit fixed effects.

Since the district border determines eligibility for incentives, it is plausible that some establishments located within district boundaries never benefit from the financial incentives. We estimate sharp discontinuity designs rather than fuzzy designs because we do not observe either: (i) the individual treated establishments that utilize specific tax incentives or (ii) the dollar amount of the incentives conditional on utilization. In other words, the best we can do in an RDD framework is to estimate a sharp design that assumes all establishments in the cultural districts receive the treatment. Of course, this is a limitation of the RDD strategy, which is why we utilize this approach as a robustness check and rely on difference-in-differences as the primary identification strategy.

We estimate the difference-in-discontinuity equations non-parametrically using a triangular kernel and the optimal data-driven bandwidth proposed by Calonico et al. (2014). This combination of kernel and bandwidth results in point estimates that satisfy a mean-squared error optimality condition (Cattaneo and Titiunik 2022).

[Table 5 here]

Estimates of the local average treatment effect near cultural district borders are reported in Table 5. Robust bias corrected standard errors are reported in parentheses (Calonico et al. 2014). Plots of the difference-in-discontinuity estimates are presented in Figure 3.

[Figure 3 here]

We find the local average treatment effects from the difference-in-discontinuity designs to be significantly different from zero using three of the four comparison groups, confirming that our estimates are not dependent on a single identification strategy. It is useful to recall that while the difference-in-differences approach captures the average difference in pre- and post-certification employment levels across all treated and comparison establishments in each respective regression, the difference-in-discontinuity treatment effects reflect differences between establishments very close to the district border, so the two sets of estimates are not directly comparable.

Our results show that establishment-level employment increases by an average of 2.6% for core cultural establishments eligible for financial incentives after certification relative to comparison establishments located within 800 meters of a district border using difference-in-differences (column 5 in Table 4). The estimated treatment effect using difference-in-discontinuity on this empirical sample is 2.8% (column 1 in Table 5). When the distance to the border is expanded out to 1600 meters, we find that establishment-level employment increases by an average of 3.0% using difference-in-differences and 2.2% using difference-in-discontinuity. The only comparison group where the difference-in-differences and difference-in-discontinuity estimators deviate in terms of statistical significance is the 1600 meter comparison group with a 400 meter buffer, shown in column 8 of Table 4 and column 4 in Table 5.²⁶

²⁶For difference-in-discontinuity specifications with comparison groups that include buffers, we remove all comparison establishments located within the buffer from the district boundary and add the buffer distance to the remaining comparison establishments. Here is a concrete example: when the buffer is 200 meters, all comparison establishments located between 0 and 200 meters are excluded to allow for potential spillover effects. Since the difference-in-discontinuity is based on establishments located very close to the border, we add the buffer distance of 200 meters from the remaining comparison establishments so that establishments located (say) 205 meters away are modeled as though they are only 5 meters away from treated establishments close to the border. Said differently, comparison establishments outside the buffer are shifted such that they are modeled *as if* they were located very close to treated establishments. This is the most direct method for making the difference-in-discontinuity approach as comparable as possible with the difference-in-differences design.

Considered jointly, the estimated treatment effects for cultural establishments range from 2.6 to 3.2% using difference-in-differences and 2.2 to 2.8% using difference-in-discontinuities. Given that the estimates rely on different identifying assumptions, there is a very high likelihood that we are estimating a credible causal effect that district financial incentives boost employment in these establishments.

5.3 Heterogeneous Effects

The overall estimated average treatment effects reported in Table 4 are a convex weighted average of potentially heterogeneous effects by state and/or financial incentive. In this section, we investigate heterogeneous effects in cultural establishment employment across states and incentives.²⁷ Given the similarities in the difference-in-differences and difference-in-discontinuity estimates presented in Sections 5.1 and 5.2, we limit our analysis in this section to difference-in-differences. We explore potential heterogeneous effects by interacting state and incentive indicator variables with the difference-in-differences variable of interest.

Since multiple states offer the same financial incentives, such as preservation credits, we are unable to isolate a state-by-incentive treatment effect because we do not have information on the specific incentives (or amounts) at the establishment level. However, this simple exploratory exercise could provide a sense of direction for future studies.

[Table 6 here]

Table 6 shows selected results from eight different regressions in two panels. Panel A interacts the difference-in-differences variable with state indicator variables to isolate the causal effect by state. Panel B interacts the difference-in-differences variable with four financial indicators: (i) =1 if the state offers a preservation tax credit, (ii) =1 for direct establishment grants, (iii) =1 for a sales tax credit, and (iv) =1 if the state has an amusement, income, or property tax credit. The amusement, income, and property tax credits are combined into a single indicator variable because Maryland is the only state to offer them, making it impossible to isolate the effect of each separately.

In terms of states, the overall employment effect identified in Table 4 is driven by the experiences of two states—Louisiana and Texas. We fail to find any evidence of an employment bump in Iowa,

²⁷We also explored heterogeneous effects among all establishments, eating and drinking establishments, retail establishments, and hotels/motels but find no evidence of any effect on these margins. These results are omitted for brevity.

Maryland, or New Mexico relative to comparison establishments. Regarding Louisiana and Texas, the interaction effects reveal that core cultural establishment employment expanded by an average of 2.3% and 4.1%, respectively, all else being equal.

Shifting to the financial incentives in Panel B, the results are less conclusive. We find, for example, that direct grants increase core cultural establishment-level employment by roughly 4% post-certification. This result is driven entirely by Texas because it is the only state that offers direct grants to eligible establishments. This result suggests that an influx of resources (as received by an establishment through a direct grant) may promote job creation and expansion among core cultural establishments. However, it is important to recall that (i) as noted in Section 2.2, grant recipients in Texas must generate at least \$50,000 and provide matching expenditures, and (ii) we do not observe which establishment receives a grant nor the amount. This may imply that care must be taken when awarding grants and leave this as an avenue for future research.

In terms of the other incentives, the point estimate on the sales tax credit indicator – which is unique to Louisiana (in our empirical sample) – averages 3.7%. It is also significant at the 10 percent level in three specifications. Louisiana also has a preservation tax credit, as do Iowa and New Mexico, and the point estimate on this interaction term is negative. Consequently, we cannot isolate clear evidence of the mechanism, or mechanisms, driving employment gains in Louisiana. The point estimates suggest that the sales tax credit (for art sold within district boundaries) may well be the channel, but this is only conjecture.

5.4 Establishment Formation

Cultural district certification may also stimulate the formation of new establishments. To explore this possibility, we modify the difference-in-differences framework and examine the number of establishments in treated and comparison groups around certified districts. Since this outcome variable is an integer, models are estimated using Poisson regression. The sample size decreases considerably because we are moving from establishment-level data to data aggregated at the cultural district and group (treated/comparison) dimensions.

We find no evidence of any effect, including heterogenous, on the number of establishments post-certification in any industry. Event study versions of these regressions also indicated no significant differences in establishment counts pre-certification between treated and comparison zones. Due to

space constraints, we report these additional regression results and event study plots in an online appendix that is available upon request.

Combining the lack of new establishment formation with the cultural employment effect (columns 5-9 in Table 4) implies that existing establishments are expanding employment after certification (relative to comparison establishments). In other words, existing cultural establishments appear to capture all program benefits. One potential explanation for this (modest) upward trend in employment is that agglomeration benefits may manifest over time as existing firms grow (Gutierrez-Posada et al. 2022; Tao et al. 2019; Bellandi et al. 2020).

5.5 Discussion

Our results indicate that existing core arts and cultural establishments experience a modest, but persistent, increase in employment following certification in Texas and Louisiana. Based on Table 3, the average cultural district in Louisiana has 21.6 cultural establishments, each with an average of 20.8 employees. The comparable figures in Texas are 71.9 and 24.6, respectively. Therefore, our estimates imply that certification generates an additional 10 jobs per district in Louisiana and 71 in Texas.²⁸

In terms of aggregate annual gains, we find that certification resulted in 970 additional jobs in Louisiana and 3,337 new jobs in Texas. Using data from the Quarterly Census of Employment and Wages for the median annual wage in cultural industries in both states –\$41,414 in Louisiana and \$51,646 in Texas– these estimates imply total annual income gains of \$40.1 million in Louisiana and \$172.3 million in Texas.²⁹

While a full accounting of the costs and benefits of cultural district programs is difficult to make, our estimates are valuable for program evaluation on the job and business creation margins. In the case of Texas, for instance, the Commission of the Arts awarded more than \$40 million in direct establishment grants between fiscal years 2012 and 2019, a sum well below our back-of-the-envelope annual income gains.³⁰ Even where we find no evidence that cultural district designation increased employment, we acknowledge that the districts may offer benefits that elude quantification, such

²⁸Consider Texas, which sees an employment increase of 4% (2.3% in Louisiana). These figures were calculated as follows: $((\text{average employees} * 1.04) - \text{average employees}) * (\text{number of establishments}) = ((24.6 * 1.04) - 24.6) * 71.9 = 70.7$ additional jobs per cultural district.

²⁹These figures use the 2018 annual average Quarterly Census on Wages and Employment (statewide) annual wages per employee for the 15 four-digit NAICS codes that encompass the 36 six-digit core cultural industry codes shown in Appendix Table A1.

³⁰<https://www.arts.texas.gov/past-grants/> (accessed: 04/19/23)

as stimulating regional economic growth by boosting cultural engagement.

6 Conclusion

State-designated cultural districts are a modern variant of place-based policies that have grown in popularity over the past two decades. Their popularity may be partly explained by the fact that these districts leverage local arts and cultural amenities to foster economic development, making them highly adaptable to the targeted area. Once a district is designated, establishments within district boundaries often become eligible to receive certain benefits (both direct and indirect). However, since state-designated cultural district programs are independently established state-level programs, the composition of benefits may vary across programs.

This study uses geocoded establishment data to explore how state designation of a cultural district affects local employment and business formation in Iowa, Louisiana, Maryland, New Mexico, and Texas. We focus on these five states because their programs offer one or more direct financial incentives to establishments located within district boundaries. Arguably, direct financial incentives increase the likelihood that cultural district designation will affect local outcomes. We explore the impact that state designation of a cultural district has on (i) overall district employment, (ii) core arts and cultural industry employment (the establishments that the policy may arguably influence the most), (iii) eating and drinking establishments, (iv) retail establishments, and (v) hotel/motel establishments. While an effect on cultural establishments may be expected from this policy, exploring the experiences of other industries that offer complementary services to residents and visitors – like restaurants and hotels – can be informative and provide evidence of additional benefits stemming from this policy.

Our main results show that state designation of a cultural district increases employment for existing cultural establishments by 2.3 to 4.0%. However, we fail to find evidence that cultural district designation affects overall district-level employment, or employment at eating and drinking establishments, retail, or hotels/motels. In other words, the policy’s effects are concentrated among arts and cultural-related establishments but do not spill over to other sectors. Furthermore, these effects appear to persist up to eight years after designation.

Since cultural district programs are independently established across states, we then proceed to investigate whether there are heterogeneous effects across states (i.e., location) and financial

incentives (i.e., the types of incentives offered for establishment located within district boundaries). Results show that the employment effects are primarily driven by the experiences of Louisiana and Texas. In terms of specific incentives, direct grants to individual establishments (used exclusively in Texas) have the largest causal effect on employment levels (establishment employment increases by roughly 4% following district certification). Our estimates are too imprecise to pinpoint whether employment gains in Louisiana are due to sales tax credits or preservation tax credits.

Nevertheless, using back-of-the-envelope calculations, we estimate that cultural district designation generates 970 and 3,337 additional core arts and cultural industry jobs in Louisiana and Texas, which equal annual income gains of \$40.1 and \$172.3 million respectively. Finally, we find no supporting evidence that district designation encourages new business formation in any sector, implying that existing cultural establishments capture all the benefits from the program’s financial incentives.

To our knowledge, this is the first study to attempt a systematic evaluation of state-designated cultural districts on local labor market outcomes using individual business establishment data spanning multiple districts and states. Our findings suggest that cultural districts can promote job development among core cultural establishments. However, it is important to keep in mind that our sample consists of the five states that offer direct financial incentives to individual establishments within district boundaries.³¹ Considering that there are thirteen other state cultural district programs operating without direct financial incentives, future research should explore whether non-financial factors tied to state designation – such as training programs or marketing assistance – could complement financial incentives and result in greater employment gains. Furthermore, the term cultural district can be broadened to include other more organic efforts to recognize an area as a hub of cultural activity. Such recognition may be driven by entrepreneurial or grassroots efforts (Galligan 2008). Whether these sets of districts are effective or not may also provide an avenue for future research. Based on our results, policymakers looking to establish a state-designated cultural district program may wish to utilize direct grants to incentivize establishments, as these appear to result in larger employment gains. If cultural districts indeed succeed at boosting employment (as suggested by the present study), increasing property values (Portillo and Wagner 2021), etc.,

³¹Nevertheless, the number of designated districts in these five states constitute 60% of all state-designated districts nationwide.

the developing area may, for example, displace current residents or artists.³² Therefore, future research may also explore the effect cultural districts have on neighborhood change and gentrification (Bereitschaft 2014).

³²However, many cultural district programs, request documentation or a discussion of affordable housing in their application. Maryland’s program guidelines state that information on the “Potential for affordable housing (rental or purchase) and studio, exhibition, or performance space” could determine eligibility for designation. Similarly, Louisiana’s guidelines state that a cultural district must “Encourage opportunity for affordable artist housing and work space.” Texas has a space asking its application to “Describe the assessment of the artists living and working in the cultural district and the amenities available to them (studio space, affordable housing, rehearsal and art making space, etc.).”

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Table 1: Cultural Districts Across States with Established Legislation

	State Authority	Number of Cultural Districts	Program Enacted	Admission & Amusement		Income Tax Credit	Preservation		Property		Sales		Establishment Grants
				Tax Exemption	Tax Credit		Tax Credit	Tax Credit	Tax Credit	Tax Credit			
AR	AR Arts Council	0	2011										
CA	CA Arts Council	14	2015										
CO	Office of Economic Development and International Trade	26	2011										
CT		Department of Economic and Community Development	1	2019									
IA*	Department of Cultural Affairs	13	2005					X					
IN	IN Arts Commission	12	2008										
KY**	KY Arts Council	6	2011										
LA	Office of Cultural Development	118	2007					X			X		
MA	MA Cultural Council	51	2010										
MD	State Arts Council	29	2001							X			
NE	NE Arts Council	0	2020		X		X						
NM	Economic Development Department	12	2007					X					
OK**	OK Arts Council	7	2013										
RI***	State Council on the Arts	9	1998								X		
SC	SC Arts Commission	9	2014				X						
TX	TX Commission on the Arts	51	2009										X
WA	WA State Arts Commission	11	2017										
WV	WV Commission on the Arts	10	2005										

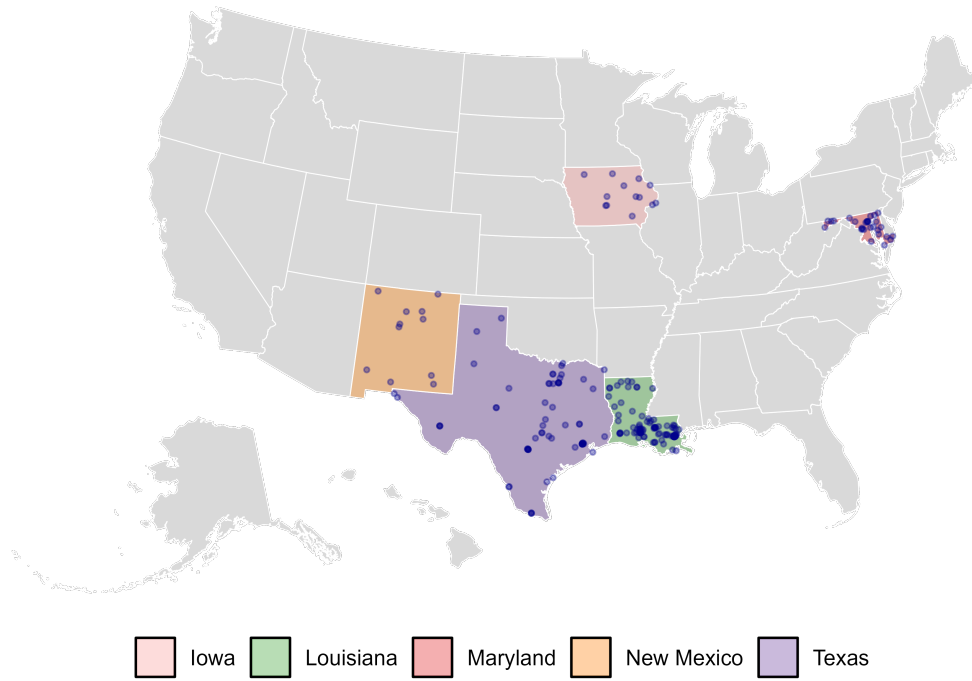
Sources: The information in this table was collected from National Assembly of State Arts Agencies [2020](#); National Assembly of State Arts Agencies [2017](#), as well as email correspondence with a NASAA research associate, and is accurate as of April 2022. Arkansas and Nebraska have legislatively established a cultural districts program but have not yet registered an operating district. The cultural district programs in Kentucky and Oklahoma are not based on legislated policy.

* The number of districts reported for Iowa in National Assembly of State Arts Agencies [2017](#) was equal to 35. However, the NASAA research associate and the Iowa cultural district's website only report 13, which is what we report in the table. Furthermore, we only obtained snapshots for these 13 districts.

** "Kentucky and Oklahoma created and actively run a program not based on legislated policy" (National Assembly of State Arts Agencies [2020](#)).

*** "Rhode Island enacted legislation in 2013 extending the sales-tax benefits of Tax-Free Arts Districts to all artists and galleries working in Rhode Island, effectively making the entire state a cultural district" (National Assembly of State Arts Agencies [2020](#)).

Figure 1: Approximate Location of State-Designated Cultural Districts



Note: This figure shows the five U.S. states that provide financial incentives to individual establishments located within state-designated cultural districts. The locations (centroids) of cultural districts within each state are shown as blue points. The points are not to scale; the objective is to show the relative positions of cultural districts within each state.

Table 2: Cultural District Program Descriptions in States Offering Financial Incentives

State	State Authority	Program Name	Program description
IA	Iowa Economic Development Authority	Cultural & Entertainment Districts	CEDs attract a diverse and engaged workforce – a key incentive for new and relocating businesses. CEDs also attract residents and tourists who support adjacent businesses such as restaurants, lodging facilities, retail stores and entertainment venues, enhancing property values, the profitability of surrounding businesses and the tax base of the area. The goals of the program are to distinguish those districts that have created unique cultural centers and prioritized visitor experiences, promote the districts and their supported activities, and provide technical assistance and professional development opportunities.
LA	Department of Culture, Recreation and Tourism	Cultural Districts	The primary goal of this initiative is to spark community revitalization based on cultural activity through tax incentives, technical assistance, and resources.
MD	Maryland State Arts Council; Department of Commerce	Arts & Entertainment Districts	The goal of the A&E Districts program is to develop, promote, and support diverse artistic and cultural centers in communities across Maryland that preserve a sense of place, provide unique local experiences, attract tourism, and spur economic revitalization and neighborhood pride. In turn, this furthers the State's goal of reinvesting in existing communities and creating places that help people, businesses, the economy, and the environment thrive as articulated in the Maryland Department of Planning's Reinvest Maryland strategy.
NM	New Mexico Economic Development Department; New Mexico Arts Commission	Arts & Cultural Districts	The Arts and Cultural Districts Program's goal is to promote the exceptional art and history of New Mexico while assisting communities in developing their cultural and artistic resources to create dynamic and economically vibrant districts. Forming Arts and Cultural Districts helps communities provide support to cultural entrepreneurs and institutions while capitalizing on the potential economic opportunity for the district as well as the town.
TX	Texas Commission on the Arts	Cultural Districts	Cultural districts are special zones that harness the power of cultural resources to stimulate economic development and community vitality. These districts can become focal points for generating businesses, attracting tourists, stimulating cultural development and fostering civic pride.

Sources: Iowa Economic Development Authority (2024), Louisiana Office of Cultural Development (2024), Maryland State Arts Council (2024), New Mexico Arts (2024) and Texas Commission on the Arts (2024).

Table 3: Establishments and Employment in State-Designated Cultural Districts

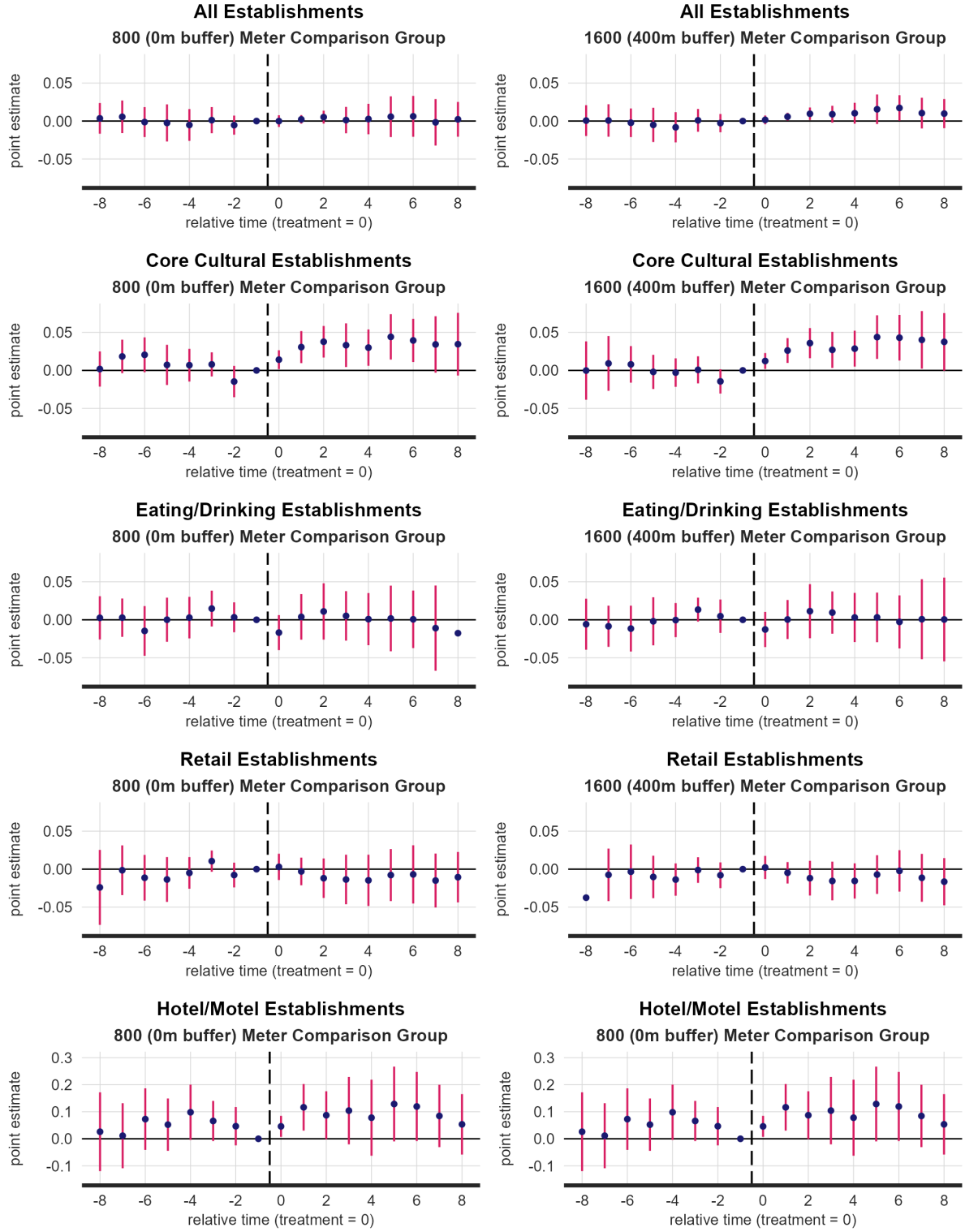
	Iowa	Louisiana	Maryland	New Mexico	Texas
Total districts	13	118	29	12	51
Districts in empirical sample	5	97	28	11	47
Median district size (km^2)	0.39	2.10	0.46	0.58	0.98
Smallest district (km^2)	0.03 The Avenues Des Moines	0.13 Welsh Cultural & Historic	0.18 Grantsville Arts and Entertainment District	0.03 Mora Arts and Cultural District	0.02 Denton
Largest district (km^2)	2.03 Valley Junction Des Moines	180.68 Bayou des Glaises	1.46 Downtown Frederick Arts and Entertainment District	2.17 Las Vegas Arts and Cultural District	33.45 Lubbock
Treated Establishments (within district boundaries)					
Mean establishments within a district	189.4	388.5	774.6	152.4	1238.2
Mean employment per establishment	6.6	6.4	10.6	8.3	8.8
Mean arts/cultural establishments	8.8	21.6	40.9	8.8	71.9
Mean employment per arts/cultural establishment	13.8	20.8	13.5	7.2	24.6
Art/cultural share of total employment	9.7%	17.2%	6.7%	4.0%	16.2%
Comparison Establishments (outside district boundaries but within 1600 meters)					
Mean number of establishments within 1600m of a district border	586.0	634.7	2113.0	703.5	2203.8
Mean employment per establishment	8.6	7.7	9.7	8.3	12.7
Mean arts/cultural establishments	32.4	27.1	137.0	44.8	129.9
Mean employment per arts/cultural establishment	16.6	29.9	15.0	16.9	23.9
Art/cultural share of total employment	10.7%	15.7%	9.9%	10.8%	11.1%

Descriptive statistics for the median, smallest, and largest districts use all cultural districts certified in a given state (as of April 2022) whether or not the district is part of our empirical analysis. Descriptive statistics for treated and comparison establishments are restricted to only certified districts that are part of the empirical analysis. The National Establishment Time Series (NETS) database covers 1990-2019, so districts certified after 2019 are excluded. Statistics on the number of establishments and employment use data from 2018. Art/cultural industries are the Arts and Cultural Production Satellite Accounts (ACPSA) defined jointly by the Bureau of Economic Analysis and National Endowment for the Arts, which are shown separately in Appendix Table A1.

Table 4: Cultural District Certification and Employment

		Dependent variable: log employees																			
D		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
		OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS
median household income		0.003 (0.009)	0.008 (0.009)	0.008 (0.008)	0.010 (0.008)	0.026** (0.011)	0.030*** (0.009)	0.032*** (0.010)	0.030*** (0.010)	-0.004 (0.010)	0.001 (0.008)	-0.003 (0.010)	0.000 (0.009)	-0.002 (0.013)	0.001 (0.010)	0.000 (0.010)	0.001 (0.011)	0.043 (0.045)	0.047 (0.043)	0.043 (0.042)	0.065 (0.047)
top marginal income tax rate		0.022*** (0.008)	0.019** (0.008)	0.020*** (0.007)	0.020*** (0.007)	0.027*** (0.009)	0.022*** (0.008)	0.020** (0.007)	0.019** (0.007)	0.009 (0.006)	0.009 (0.008)	0.007 (0.006)	0.003 (0.006)	0.020 (0.012)	0.015 (0.011)	0.016 (0.011)	0.015 (0.011)	-0.005 (0.011)	-0.019 (0.025)	-0.013 (0.028)	-0.013 (0.029)
unemployment rate		0.005*** (0.001)	0.005*** (0.001)	0.005*** (0.001)	0.005*** (0.001)	0.006*** (0.001)	0.005*** (0.001)	0.005*** (0.001)	0.005*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.004*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)	0.012*** (0.004)	0.012*** (0.003)	0.011*** (0.003)	0.009*** (0.003)
median house price		-0.010*** (0.002)	-0.010*** (0.001)	-0.010*** (0.001)	-0.010*** (0.001)	-0.008*** (0.003)	-0.009*** (0.002)	-0.010*** (0.002)	-0.010*** (0.002)	-0.029*** (0.002)	-0.028*** (0.002)	-0.028*** (0.001)	-0.029*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)	-0.006*** (0.002)	0.000 (0.008)	-0.003 (0.006)	-0.003 (0.006)	-0.001 (0.005)
trend personal income growth		0.000 (0.011)	-0.003 (0.011)	-0.004 (0.010)	-0.006 (0.009)	0.022*** (0.006)	0.013** (0.006)	0.013** (0.006)	0.012** (0.006)	0.022** (0.013)	0.017 (0.011)	0.013 (0.010)	0.016 (0.011)	0.016 (0.011)	-0.013 (0.013)	-0.019 (0.013)	-0.022* (0.013)	-0.006 (0.006)	-0.027 (0.037)	-0.050 (0.044)	-0.030 (0.037)
average labor intensity		0.012 (0.017)	0.017 (0.016)	0.016 (0.014)	0.014 (0.013)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	-0.006*** (0.008)	0.003 (0.003)	0.003 (0.003)	0.002 (0.003)	0.003 (0.003)
airline travelers		-0.005 (0.168)	-0.009 (0.148)	-0.055 (0.150)	-0.087 (0.147)	-0.054 (0.317)	-0.175 (0.296)	-0.276 (0.322)	-0.242 (0.348)	-0.091 (0.300)	-0.035 (0.226)	-0.092 (0.286)	-0.092 (0.286)	0.000 (0.268)	0.004 (0.250)	0.025 (0.240)	0.084 (0.256)	-0.850 (0.858)	-0.321 (0.785)	-0.483 (0.806)	-0.232 (1.004)
empowerment/renewal/enterprise zone		-4.717 (3.234)	-2.740 (2.653)	-2.538 (2.632)	-2.053 (2.640)	8.221* (4.719)	7.774 (6.204)	6.741 (5.714)	7.562 (5.884)	16.806** (7.985)	17.738** (6.563)	20.773*** (6.142)	20.585*** (5.615)	-3.379 (5.811)	-1.831 (4.695)	-4.916 (5.604)	-5.988 (5.654)	43.092 (36.706)	41.037* (24.260)	37.413 (23.255)	33.113 (23.057)
cultural human capital		0.000 (0.007)	-0.005 (0.010)	-0.004 (0.010)	-0.004 (0.011)	-0.006 (0.059)	-0.007 (0.048)	-0.005 (0.044)	-0.012 (0.051)	0.058** (0.023)	0.062** (0.027)	0.054* (0.028)	0.053* (0.027)	-0.042 (0.031)	-0.089** (0.030)	-0.081** (0.032)	-0.077* (0.043)	-0.271** (0.130)	-0.126 (0.123)	-0.141 (0.133)	-0.127 (0.133)
N		0.045 (0.389)	-0.041 (0.310)	-0.056 (0.297)	-0.247 (0.280)	-0.655 (1.150)	-1.186 (0.763)	-1.231 (0.754)	-1.938** (0.873)	0.660 (1.163)	0.523 (0.790)	0.564 (0.934)	0.588 (0.944)	-1.570*** (0.565)	-1.675*** (0.471)	-1.209* (0.644)	-1.433* (0.721)	-8.447*** (1.434)	-7.809*** (1.755)	-7.073*** (2.133)	-13.060*** (4.308)
Establishment FE		2.042,235 (yes)	2.790,387 (yes)	2.516,170 (yes)	2.266,657 (yes)	114,012 (yes)	156,699 (yes)	142,219 (yes)	128,042 (yes)	94,322 (yes)	127,052 (yes)	114,835 (yes)	104,249 (yes)	69,559 (yes)	95,631 (yes)	87,717 (yes)	80,978 (yes)	14,011 (yes)	17,835 (yes)	16,346 (yes)	14,673 (yes)
Year FE		yes (All)	yes (All)	yes (All)	yes (All)	yes (Cultural)	yes (Cultural)	yes (Cultural)	yes (Cultural)	yes (Eating/Drinking)	yes (Eating/Drinking)	yes (Eating/Drinking)	yes (Eating/Drinking)	yes (Retail)	yes (Retail)	yes (Retail)	yes (Retail)	yes (Hotel/Motel)	yes (Hotel/Motel)	yes (Hotel/Motel)	yes (Hotel/Motel)
Establishments		109,347 (yes)	109,347 (yes)	109,347 (yes)	109,347 (yes)	6,437 (yes)	6,437 (yes)	6,437 (yes)	6,437 (yes)	4,995 (yes)	4,995 (yes)	4,995 (yes)	4,995 (yes)	4,328 (yes)	4,328 (yes)	4,328 (yes)	4,328 (yes)	752 (yes)	752 (yes)	752 (yes)	752 (yes)
Treated Establishments		105,867 (yes)	186,734 (yes)	156,839 (yes)	129,322 (yes)	5,995 (yes)	6,437 (yes)	6,437 (yes)	6,437 (yes)	4,256 (yes)	7,551 (yes)	6,295 (yes)	5,193 (yes)	3,122 (yes)	5,874 (yes)	4,975 (yes)	4,227 (yes)	572 (yes)	990 (yes)	839 (yes)	666 (yes)
Comparison Establishments		800m (yes)	1600m (yes)	1600m (yes)	1600m (yes)	800m (yes)	1000m (yes)	1600m (yes)	1600m (yes)	800m (yes)	1600m (yes)	1600m (yes)	1600m (yes)	800m (yes)	1600m (yes)	1600m (yes)	1600m (yes)	800m (yes)	1600m (yes)	1600m (yes)	1600m (yes)
Comparison Threshold		0m (yes)	0m (yes)	200m (yes)	400m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)
Comparison Buffer		0m (yes)	0m (yes)	200m (yes)	400m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)	0m (yes)

Figure 2: Event Study Formulations: Employment



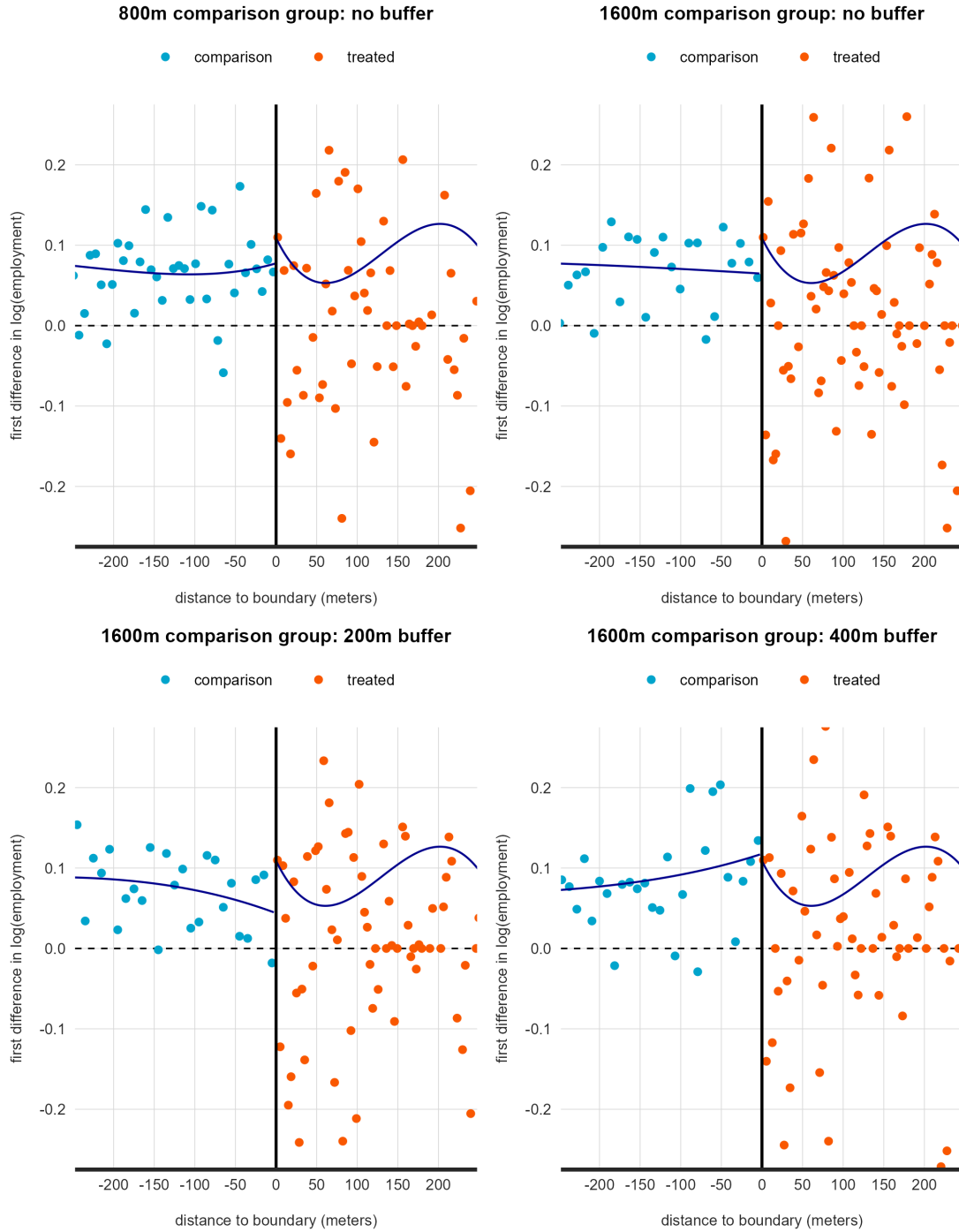
Note: This figure plots selected event study formulations of the difference-in-differences regressions shown in Table 4. Relative time period -1 is the omitted time indicator from the regressions. Error bars denote 95% confidence intervals.

Table 5: Difference-in-Discontinuity Estimates

<i>Dependent variable:</i>				
core cultural establishments				
	(1)	(2)	(3)	(4)
point estimate	0.028** (0.012)	0.022** (0.010)	0.022** (0.011)	0.008 (0.013)
Comparison Threshold	800m	1600m	1600m	1600m
Comparison Buffer	0m	0m	200m	400m
Establishments	Core Cultural	Core Cultural	Core Cultural	Core Cultural

This table presents the local average treatment effect estimate from a sharp regression discontinuity design using each establishment's distance to the closest cultural district boundary as the running variable. Robust bias-corrected standard errors reported in parentheses. Sample includes districts in Iowa, Louisiana, Maryland, New Mexico, and Texas. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. If the comparison buffer is greater than zero, then comparison establishments within the buffer zone from a district boundary are excluded from the comparison group.

Figure 3: Difference-in-Discontinuity Robustness Checks



Note: This figure plots the difference-in-discontinuity robustness checks for employment in core cultural establishments. Figures are from a sharp first-difference regression discontinuity using the establishment's distance to the boundary (in meters) as the running variable. The range of the x-axis in the figures have been truncated for better visibility.

Table 6: District Certification and Employment: Heterogenous Effects

	<i>Dependent variable:</i> log employees			
	(1) OLS	(2) OLS	(3) OLS	(4) OLS
<i>Panel A: Heterogenous effects by state</i>				
D * IA	-0.016 (0.019)	-0.018 (0.019)	-0.017 (0.019)	-0.019 (0.019)
D * LA	0.018* (0.011)	0.024** (0.009)	0.026** (0.010)	0.024** (0.010)
D * MD	0.003 (0.023)	0.010 (0.023)	0.013 (0.024)	0.010 (0.025)
D * NM	0.011 (0.015)	0.018 (0.014)	0.024 (0.016)	0.020 (0.017)
D * TX	0.040** (0.015)	0.042*** (0.013)	0.043*** (0.013)	0.041*** (0.012)
<i>Panel B: Heterogenous effects by financial incentive</i>				
D * preservation credit	-0.016 (0.019)	-0.018 (0.019)	-0.018 (0.019)	-0.019 (0.019)
D * direct grant	0.040** (0.015)	0.042*** (0.013)	0.043*** (0.013)	0.041*** (0.012)
D * sales tax credit	0.034 (0.024)	0.041* (0.023)	0.043* (0.022)	0.042* (0.023)
D * amusement, income, property tax	0.003 (0.023)	0.010 (0.023)	0.013 (0.024)	0.010 (0.024)
N	114,012	156,699	142,219	128,042
Establishment FE	yes	yes	yes	yes
Year FE	yes	yes	yes	yes
Establishments	Core Cultural	Core Cultural	Core Cultural	Core Cultural
Treated Establishments	6,437	6,437	6,437	6,437
Comparison Establishments	5,995	10,739	9,117	7,518
Comparison Threshold	800m	1600m	1600m	1600m
Comparison Buffer	0m	0m	200m	400m

This table presents the effect of state cultural district certification on establishment-level employment. The results shown in Panel A and Panel B are from different regressions showing only the parameters of interest. The full set of fixed effects and covariates included in Table 4 are omitted from this table for brevity. Establishments within cultural districts are compared to establishments just outside of the district boundaries. D denotes the difference-in-differences parameter of interest. Sample includes districts in Iowa, Louisiana, Maryland, New Mexico, and Texas. Comparison establishments are specified in meters. Standard errors are shown in parentheses and are clustered at the cultural district and metropolitan statistical area dimensions. Establishment and year by establishment size fixed effects are included but not reported. *** denotes significance at the 1 percent level, ** at the 5 percent level, and * at the 10 percent level. If comparison buffer is greater than zero, then comparison establishments within the buffer zone from a district boundary are excluded from the comparison group.

A Appendix

Table A1: ACPSA Cultural NAICS Industries

ACPSA description	ACPSA 2 digit	NAICS code	NAICS title
Performing arts	35	711110	Theater companies and dinner theaters
		711120	Dance companies
		711130	Musical groups and artists
		711190	Other performing arts companies
Independent artists, writers, and performers	38	711510	Independent artists, writers, and performers
Agents/managers for artists	37	711410	Agents and managers for artists, athletes, entertainers, and other public figures
Promoters of performing arts and similar events	36	711310	Promoters of performing arts, sports, and similar events with facilities
		711320	Promoters of performing arts, sports, and similar events without facilities
Museums	39	712110	Museums
		712120	Historical sites
		712130	Zoos and botanical gardens
		712190	Nature parks and other similar institutions
Advertising	30	541810	Advertising agencies
		541820	Public relations agencies
		541830	Media buying agencies
		541840	Media representatives
		541850	Display advertising
		541860	Direct mail advertising
		541890	Other services related to advertising
Architectural services	23	541310	Architectural services
Landscape architectural services	24	541320	Landscape architectural services
Interior design services	25	541410	Interior design services
Industrial design services	26	541420	Industrial design services
Graphic design services	27	541430	Graphic design services
All other design services	28	541490	Other specialized design services
Computer systems design	29	541511	Computer systems design and related services
		541519	Other computer related services
Photography and finishing services	31	541921	Photography studios, portrait
		541922	Commercial photography
		812921	Photofinishing laboratories (except one-hour)
Fine arts education	34	611610	Fine arts schools
Education services	33	611110	Elementary and secondary schools
		611210	Junior colleges
		611310	Colleges, universities, and professional schools
		611511	Cosmetology and barber schools
		611519	Other technical and trade schools

Notes: Arts and Cultural Production Satellite Account (ACPSA) is a partnership between the National Endowment for the Arts (NEA) Office of Research & Analysis and the U.S. Bureau of Economic Analysis (BEA). This concordance showing the ACPSA codes mapped to North American Industrial Classification (NAICS) codes is available here: <https://apps.bea.gov/regional/xls/ListofARTSIndustries.xlsx> [accessed: May 27, 2024].